Database Systems
Practice Examination

Preamble

Please read the following instructions carefully:

- This exam has seven questions and 13 pages. Make sure your copy contains all exercises and all questions. The following table summarizes the questions in this exam and their weights.

Question	**Points**
1. Relational Algebra	10
2. SQL	20
3. ER Modeling	16
4. Design Theory	14
5. Physical Design	12
6. Query Processing and Optimization	14
7. Transactions, Concurrency Control, and Recovery	14
Total	**100**

- Read the text of each exercise carefully before solving it.
- If you get stuck on an exercise, proceed with another one and come back later.
- If your solution is based on an assumption that you have made, please add a comment for clarification.
- Aids are allowed during the exam, i.e., you can consult aids such as books, notes, etc.
- As a reference, we recommend the course material (book, slides, exercises, etc.) as it complies with the notation used in the exam.
- It is NOT allowed to seek help from or consult others (human beings, artificial intelligence, etc.) during the exam.
- Good luck!

1 Relational Algebra (10 points)

Consider the following relations capturing information about a library system with books, authors, members, and borrowings:

```
author(aid, name, birthYear, nationality)
book(bid, title, isbn, publishYear, aid -> author)
member(mid, name, email, joinDate)
borrowing(mid -> member, bid -> book, borrowDate, returnDate, fine)
```

where the primary keys are underlined and the foreign keys are denoted by arrows.

Fill in the missing information that should go into the boxes to make the query compute the required information. Note that your solution should not output additional information.

To enter the operation symbols into the answer sheet, please use the following replacements:

Operation symbol	What to write in the answer sheet
ρ	rho
γ	gamma
Π	pi
σ	sigma
$\bowtie$	join
$\times$	x

1.1 (4 points)

Find all members who have borrowed books written by authors with nationality 'British'. For each such member, list their mid and name.

$$\boxed{\text{Box 1}}_{\text{mid, name}} \ (\text{member} \ \boxed{\text{Box 2}} \ \text{borrowing} \ \boxed{\text{Box 3}} \ \text{book} \ \boxed{\text{Box 4}}_{\text{book.aid = author.aid}} \ (\boxed{\text{Box 5}}_{\text{nationality = 'British'}} \ (\text{author})))$$

1.2 (6 points)

Find all authors who have written at least one book that has never been borrowed (no borrowing record exists for that book). For each such author, list their aid.

$$\boxed{\text{Box 6}}_{\text{aid}} \ (\text{author} \ \boxed{\text{Box 7}}_{\text{author.aid = book.aid}} \ (\boxed{\text{Box 8}}_{\text{bid}} \ (\text{book}) \ \boxed{\text{Box 9}} \ \boxed{\text{Box 10}}_{\text{bid}} \ (\text{borrowing})))$$

2 SQL (20 points)

2.1 Evaluate whether the following statements about SQL are true or false.

2.1.1 (2 points)

In SQL, COUNT(*) counts all rows including those with NULL values, while COUNT(column) counts only non-NULL values in that column.

(a) True (b) False

2.1.2 (2 points)

The condition `age > ALL (SELECT age FROM student WHERE major = 'CS')` is TRUE when the subquery returns an empty result set.

(a) True (b) False

2.2 Consider again the relational schema from Question 1:

```
author(aid, name, birthYear, nationality)
book(bid, title, isbn, publishYear, aid -> author)
member(mid, name, email, joinDate)
borrowing(mid -> member, bid -> book, borrowDate, returnDate, fine)
```

where the primary keys are underlined and the foreign keys are denoted by arrows.

Answer the following questions:

2.2.1 (4 points)

Which of the following queries correctly finds all members who have borrowed books with fines greater than 50? For each such member, list their mid and name. Choose **only one** option:

(a) SELECT m.mid, m.name
FROM member m, borrowing b
WHERE fine > 50;

(b) SELECT DISTINCT m.mid, m.name
FROM member m JOIN borrowing b ON m.mid = b.mid
WHERE b.fine > 50;

(c) SELECT m.mid, m.name
FROM member m
WHERE m.mid IN (SELECT mid FROM borrowing WHERE fine > 50);

(d) Both (b) and (c) are correct.

(e) SELECT m.mid, m.name
FROM member m NATURAL JOIN borrowing
HAVING fine > 50;

(f) None of the above

2.2.2 (4 points)

Which of the following queries correctly finds nationalities that have more than 3 authors? For each such nationality, list the nationality and the count. Choose **only one** option:

(a) SELECT nationality, COUNT(*)
FROM author
WHERE COUNT(*) > 3
GROUP BY nationality;

(b) SELECT nationality, COUNT(aid)
FROM author
GROUP BY nationality
HAVING COUNT(*) > 3;

(c) SELECT nationality, COUNT(*)
FROM author
GROUP BY nationality
WHERE COUNT(*) > 3;

(d) SELECT nationality
FROM author
HAVING COUNT(*) > 3;

(e) Both (a) and (b) are correct.

(f) None of the above

2.2.3 (4 points)

Fill in the missing information that should go into the boxes of the following query: find all books that have been borrowed by at least one member who joined after '2022-01-01'. For each such book, list bid and title.

```
SELECT DISTINCT b.bid, b.title
FROM book b
WHERE [Box 1] (
    SELECT br.bid
    FROM borrowing br JOIN member m ON [Box 2]
    WHERE [Box 3]
);
```

2.2.4 (4 points)

Consider the following query that finds members who have never borrowed any book:

```
SELECT m.mid, m.name
FROM member m LEFT JOIN borrowing br ON m.mid = br.mid
WHERE br.mid IS NULL;
```

Which of the following statements is **correct**? Choose **one or more** options:

(a) This query uses an outer join to preserve all members.

(b) The WHERE clause filters for members with no matching borrowing records.

(c) This query is equivalent to: `SELECT mid, name FROM member WHERE mid NOT IN (SELECT mid FROM borrowing)`.

(d) If we replace LEFT JOIN with INNER JOIN, the query would return all members who have borrowed books.

(e) The condition `br.mid IS NULL` checks for members without any borrowing records.

(f) All of the above except (d).

3 ER Modeling (16 points)

3.1 Evaluate whether the following statements about ER model concepts are true or false.

3.1.1 (2 points)

In an ER diagram, a relationship can have descriptive attributes.

(a) True (b) False

3.1.2 (2 points)

A weak entity set can exist in a database without being related to its owner entity.

(a) True (b) False

3.1.3 (2 points)

Composite attributes can be further divided into smaller sub-attributes.

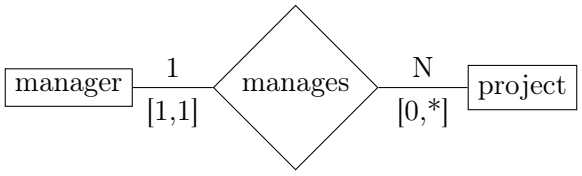
(a) True (b) False

3.2 (4 points)

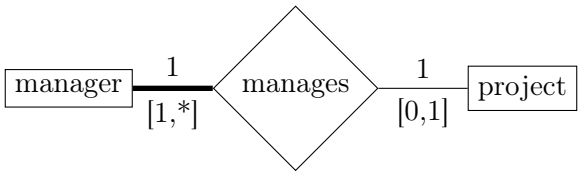
Consider the relationship *manages* where

- each *project* can be managed by at most one manager (partial participation)
- each *manager* must manage at least one project (total participation)

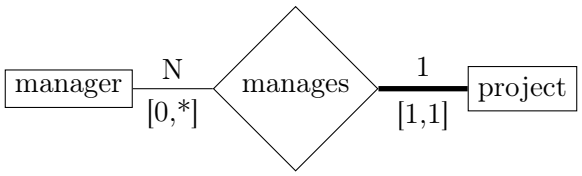
Which of the following ER diagram representations is **correct**? Choose **only one** option:



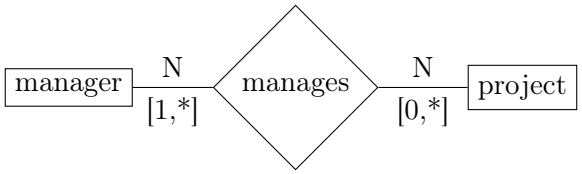
(a)



(b)



(c)

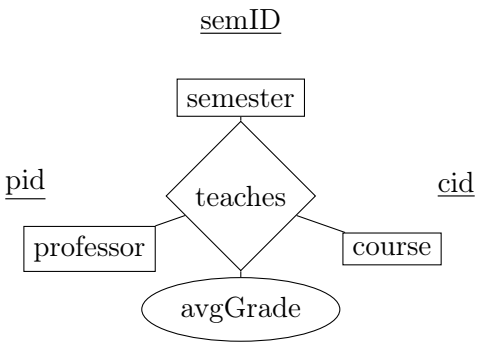


(d)

(e) None of the above

3.3 (6 points)

Consider the following ER diagram showing a ternary relationship among *professor*, *course*, and *semester*:



A professor can teach multiple courses in different semesters. Each (professor, course, semester) combination has an average grade. Which combination of relations should be created? Choose **one or more** options:

(a) professor(pid, name, office)

(b) course(cid, title, credits)

(c) semester(semID, year, term)

(d) teaches(pid → professor, cid → course, semID → semester, avgGrade)

(e) teaches(pid → professor, cid → course, semID → semester, avgGrade)

(f) teaches(pid → professor, cid → course, semID → semester, avgGrade)

(g) avgGrade(pid → professor, cid → course, semID → semester, grade)

(h) None of the above

4 Design Theory (14 points)

4.1 Evaluate whether the following statements about design theory are true or false.

4.1.1 (2 points)

The closure of an attribute set includes all attributes that can be functionally determined by it.

(a) True (b) False

4.1.2 (2 points)

A relation is in BCNF if and only if it is in 3NF.

(a) True (b) False

4.2 Consider a relation schema $R = (P, Q, R, S, T)$ with the following functional dependencies (FDs):

$$\begin{aligned} PQ &\rightarrow R \\ R &\rightarrow ST \\ S &\rightarrow Q \end{aligned}$$

Answer the following questions:

4.2.1 (4 points)

Which of the following attributes (sets) can be a **superkey** of R? Choose **one or more** options:

(a) P

(b) Q

(c) R

(d) S

(e) PQ

(f) PR

(g) PS

(h) RS

(i) PQR

(j) PQRST

(k) None of the above

4.2.2 (3 points)

Which of the following statements is correct about the normalization status of relation R? Choose **only one** option:

(a) The relation R is in BCNF and 3NF.

(b) The relation R is in BCNF but not in 3NF.

(c) The relation R is in 3NF but not in BCNF.

(d) The relation R is not in BCNF or 3NF.

(e) None of the above

4.2.3 (3 points)

Which of the following decompositions of the relational schema $R = (P, Q, R, S, T)$ is **lossless**? Choose **one or more** options:

(a) $R_1 = (P, Q, R)$ and $R_2 = (P, S, T)$

(b) $R_1 = (P, Q, R)$ and $R_2 = (R, S, T)$

(c) $R_1 = (P, R, S)$ and $R_2 = (Q, R, T)$

(d) $R_1 = (S, Q, R)$ and $R_2 = (P, Q, T)$

(e) None of the above

5 Physical Design (12 points)

5.1 Evaluate whether the following statements about B+ Trees are true or false.

5.1.1 (2 points)

In a B+ Tree, internal nodes contain copies of keys that also appear in leaf nodes.

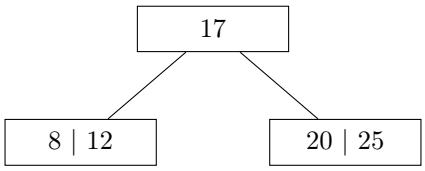
(a) True (b) False

5.1.2 (2 points)

For a B+ Tree with degree $d = 4$, a leaf node can contain a maximum of 3 keys.

(a) True (b) False

5.2 Consider the following B+ Tree with degree $d = 3$.



Visual representation. Actual leaf structure:

- Leaves: [6* 8*] | [12* 15*] | [17* 20*] | [25* 30*]
- Root: contains key 17 with 2 pointers
- Internal nodes: Left child contains keys [8, 12]; Right child contains keys [20, 25]

Please make the following assumptions:

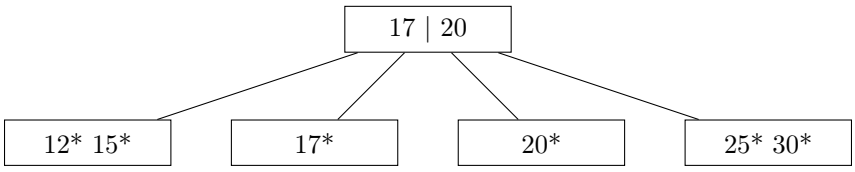
- The left pointer of a key k in a non-leaf node leads towards keys less than k, while the right pointer leads towards keys greater than or equal to k.
- A leaf node underflows when the number of keys goes below $\lceil \frac{d-1}{2} \rceil$.
- An internal node underflows when the number of pointers goes below $\lceil \frac{d}{2} \rceil$.

Answer the following questions:

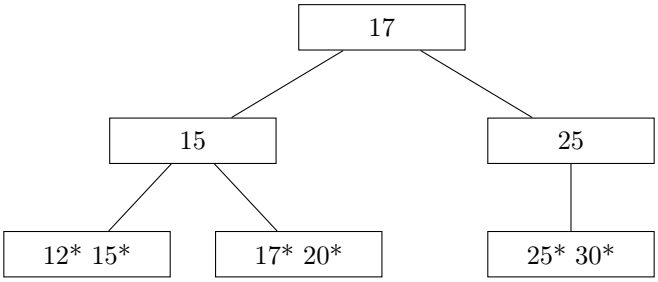
5.2.1 (4 points)

Delete 6*, then delete 8*. What is the resulting tree? Choose **only one** option:

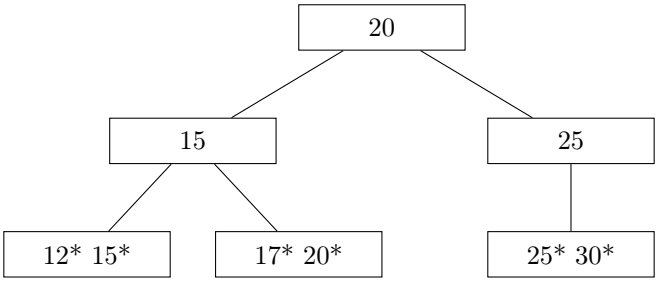
(a)



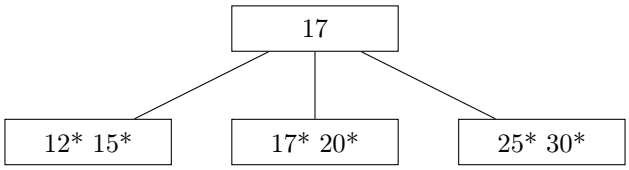
(b)



(c)



(d)

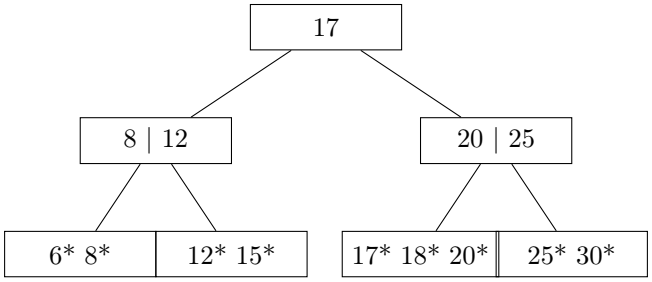


(e) None of the above

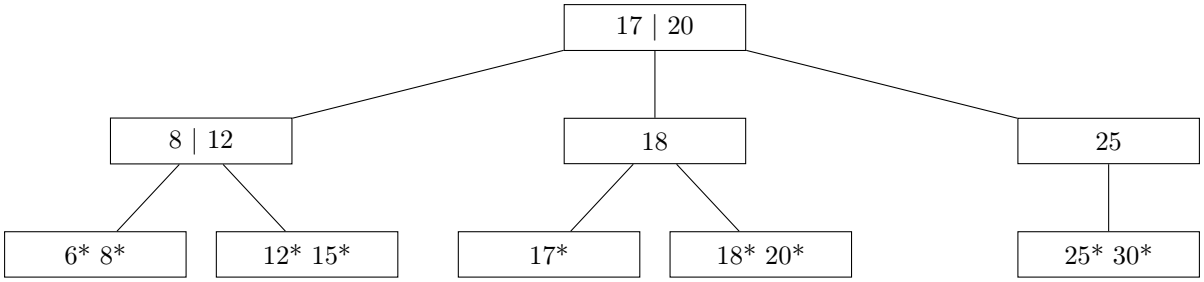
5.2.2 (4 points)

Consider the original tree (ignore operations in 5.2.1). Insert 18* into the tree. What is the resulting tree? Choose **only one** option:

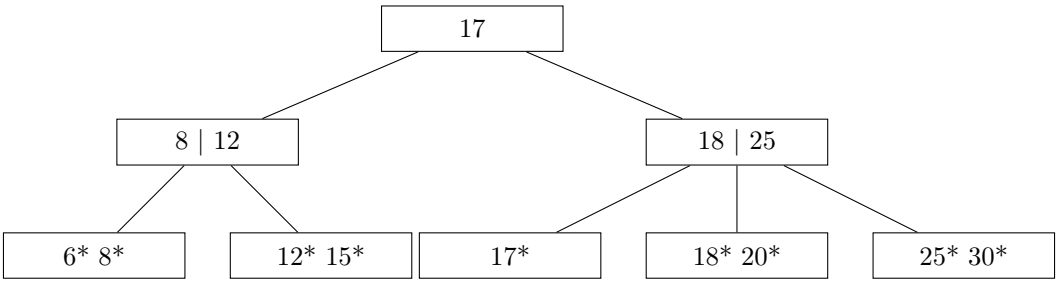
(a)



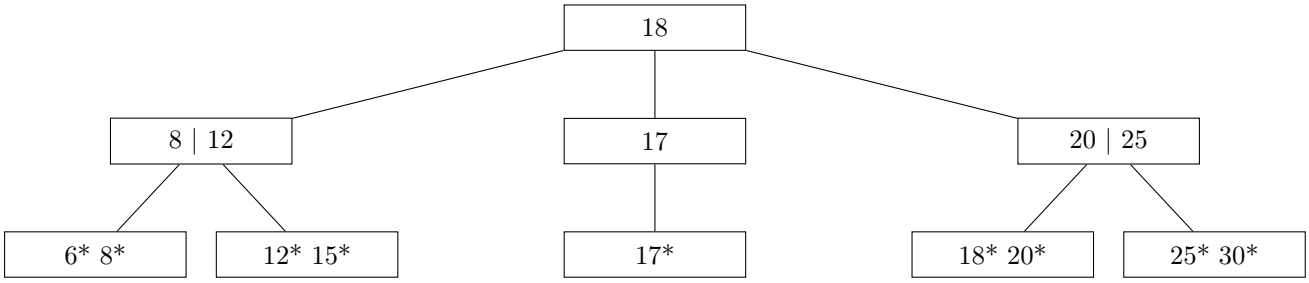
(b)



(c)



(d)



(e) None of the above

6 Query Processing and Optimization (14 points)

Consider a database containing relations s_1 and s_2 with the following properties: s_1 has 60,000 tuples, s_2 has 300,000 tuples, 30 tuples of s_1 fit on one page, and 15 tuples of s_2 fit on one page. Assume the computer has 250 buffer pages.

Answer the following questions:

6.1 (2 points)

Suppose you need to sort s_1 using the 2-Way External Merge Sort algorithm. How many passes are needed in Phase 2? Choose **only one** option:

(a) $\lceil \log_2 60,000 \rceil$

(b) $\lceil \log_2 2,000 \rceil$

(c) $\lceil \log_{249} \lceil 2,000/250 \rceil \rceil$

(d) $\lceil \log_{249} \lceil 60,000/250 \rceil \rceil$

(e) None of the above

6.2 (4 points)

Suppose you need to sort s_2 using the Multi-Way External Merge Sort algorithm. Which of the following statements is **correct**? Choose **one or more** options:

(a) The number of pages for s_2 is 20,000.

(b) The I/O cost in Phase 1 is 40,000 page transfers.

(c) The number of initial sorted runs created in Phase 1 is $\lceil 20,000/250 \rceil = 80$.

(d) The number of passes needed in Phase 2 is $\lceil \log_{249} 80 \rceil = 1$.

(e) The total I/O cost for sorting is $2 \times 20,000 \times (1 + 1) = 80,000$ page transfers.

(f) None of the above

6.3 (4 points)

Suppose you need to join s_1 and s_2 using the block-oriented nested-loop join algorithm with s_1 as outer. Which of the following statements is **correct**? Choose **only one** option:

(a) The number of pages for s_1 is 2,000 and for s_2 is 20,000.

(b) The total I/O cost is $2,000 + 2,000 \times 20,000 = 40,002,000$ page transfers.

(c) If we use s_2 as outer instead, the cost would be $20,000 + 20,000 \times 2,000 = 40,020,000$ page transfers.

(d) Both (a) and (b) are correct.

(e) None of the above

6.4 (4 points)

Given a relation $s(X, Y, Z)$ with $n_s = 5000$, $V(X, s) = 200$, $V(Y, s) = 250$, and $V(Z, s) = 50$, where n_s is the number of tuples in relation s and $V(X, s)$ is the number of distinct values that appear in s for attribute X.

Assume the range of values for attribute X is [100, 300], for attribute Y is [0, 500], and for attribute Z is [50, 150]. The values are uniformly distributed.

Which of the following statements is **correct**? Choose **one or more** options:

(a) The size estimation of the selection operation $\sigma_{Z=100}(s)$ is 100.

(b) The size estimation of the selection operation $\sigma_{X=200}(s)$ is 25.

(c) The size estimation of the selection operation $\sigma_{Y<250}(s)$ is 2500.

(d) The size estimation of the selection operation $\sigma_{X<200}(s)$ is 2500.

(e) The size estimation of the selection operation $\sigma_{Z>100}(s)$ is 2500.

(f) None of the above

7 Transactions, Concurrency Control, and Recovery (14 points)

7.1 Evaluate whether the following statements are true or false.

7.1.1 (2 points)

The REDO phase in log-based recovery processes the log in forward direction (from beginning to end).

(a) True (b) False

7.1.2 (2 points)

Deadlock can occur when two or more transactions are waiting for locks held by each other.

(a) True (b) False

7.2 We consider three transactions T1, T2, and T3 consisting of the following actions:

```
T1: R(M) W(N) C
T2: W(M) R(P) W(P) C
T3: R(N) W(M) R(P) C
```

R(X) represents reading object X, W(X) represents writing object X, and C represents Commit.

Consider the following two schedules:

Schedule (1):

Transaction	Operations
T1	R(M) W(N) C
T2	W(M) R(P) W(P) C
T3	R(N) W(M) R(P) C

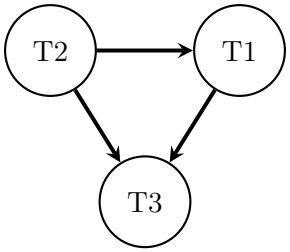
Schedule (2):

Transaction	Operations
T1	R(M) W(N) C
T2	W(M) R(P) W(P) C
T3	R(N) W(M) R(P) C

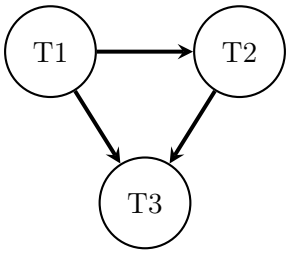
Answer the following questions:

7.2.1 (3 points)

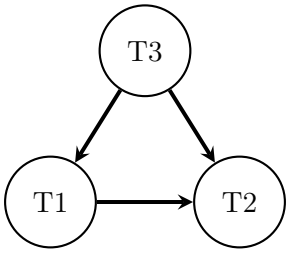
Which precedence graph of Schedule (1) is **correct**? Choose **one** option:



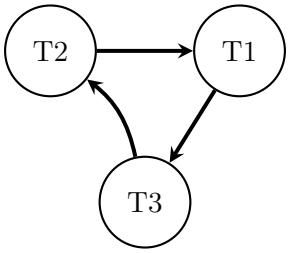
(a)



(b)



(c)



(d)

(e) None of the above

7.2.2 (4 points)

Which of the following statements about Schedule (1) is **correct**? Choose **one or more** options:

(a) Schedule (1) is a serial schedule.

(b) Schedule (1) is a conflict serializable schedule.

(c) Schedule (1) is a serializable schedule.

(d) The equivalent serial schedule of Schedule (1) can be T2, T1, T3.

(e) The equivalent serial schedule of Schedule (1) can be T1, T2, T3.

(f) None of the above

7.2.3 (3 points)

Analyze Schedule (2). Which type(s) of schedule does Schedule (2) belong to? Choose **one or more** options:

(a) Serial schedule

(b) Serializable schedule

(c) Conflict serializable schedule

(d) None of the above

ANSWERS SECTION

Question 1: Relational Algebra

1.1 (4 points)

Box 1: pi

Box 2: join

Box 3: join

Box 4: join

Box 5: sigma

Complete answer: $\Pi_{\text{mid, name}}(\text{member} \bowtie \text{borrowing} \bowtie \text{book} \bowtie_{\text{book.aid = author.aid}} (\sigma_{\text{nationality = 'British'}}(\text{author})$

1.2 (6 points)

Box 6: pi

Box 7: join

Box 8: pi

Box 9: - (minus)

Box 10: pi

Complete answer: $\Pi_{\text{aid}}(\text{author} \bowtie_{\text{author.aid = book.aid}} (\Pi_{\text{bid}}(\text{book}) - \Pi_{\text{bid}}(\text{borrowing})))$

Explanation: Find books that exist but have never been borrowed (set difference), then join with authors to get their aids.

Question 2: SQL

2.1.1 (2 points)

Answer: (a) True

COUNT(*) counts all rows regardless of NULL values. COUNT(column) only counts non-NULL values in that specific column.

2.1.2 (2 points)

Answer: (a) True

When the subquery returns an empty set, `age > ALL (empty set)` is vacuously true - any value is greater than all zero values.

2.2.1 (4 points)

Answer: (d) Both (b) and (c) are correct

(b) uses DISTINCT with JOIN and proper WHERE clause. (c) uses subquery with IN. Both correctly find members with fines > 50. Option (a) is missing the join condition between m and b.

2.2.2 (4 points)

Answer: (b)

Correct: GROUP BY nationality, then HAVING COUNT(*) > 3 filters groups. Cannot use COUNT in WHERE (option a). Option (c) has wrong order (WHERE after GROUP BY). Option (d) is missing GROUP BY.

2.2.3 (4 points)

Box 1: b.bid IN

Box 2: br.mid = m.mid

Box 3: m.joinDate > '2022-01-01'

Complete query:

```
SELECT DISTINCT b.bid, b.title
FROM book b
WHERE b.bid IN (
    SELECT br.bid
    FROM borrowing br JOIN member m ON br.mid = m.mid
    WHERE m.joinDate > '2022-01-01'
);
```

2.2.4 (4 points)

Answer: (f) All of the above except (d)

(a) TRUE: LEFT JOIN preserves all members. (b) TRUE: WHERE br.mid IS NULL filters for members without borrowing records. (c) TRUE: equivalent to NOT IN approach. (e) TRUE: IS NULL checks for no matching records. (d) FALSE: INNER JOIN with WHERE br.mid IS NULL would return empty set (contradiction).

Question 3: ER Modeling

3.1.1 (2 points)

Answer: (a) True

Relationships can have descriptive attributes (e.g., a "works_in" relationship can have a "since" attribute).

3.1.2 (2 points)

Answer: (b) False

Weak entities have existence dependency on their owner entity - they cannot exist without it.

3.1.3 (2 points)

Answer: (a) True

Composite attributes can be subdivided (e.g., Address can be broken into Street, City, Zip).

3.2 (4 points)

Answer: (b)

Manager [1,*] with double line (total participation) means each manager must manage at least one project. Project [0,1] means each project can be managed by at most one manager.

3.3 (6 points)

Answer: (a), (b), (c), (d)

For a ternary relationship, we need:

- All three entity relations: (a) professor, (b) course, (c) semester
- The relationship relation: (d) teaches with all three foreign keys as composite primary key, plus the avgGrade attribute

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Question 4: Design Theory

4.1.1 (2 points)

Answer: (a) True

The closure α^+ contains all attributes functionally determined by α.

4.1.2 (2 points)

Answer: (b) False

BCNF is stricter than 3NF. All BCNF relations are in 3NF, but not all 3NF relations are in BCNF.

4.2.1 (4 points)

Answer: (e), (f), (g), (j)

Computing closures:

- $P^+ = \{P\}$ - not a superkey
- $(PQ)^+ = \{P, Q, R, S, T\}$ (using $PQ \to R$, then $R \to ST$) - **superkey!**
- $(PR)^+ = \{P, R, S, T, Q\}$ (using $R \to ST$, $S \to Q$) - **superkey!**

- $(PS)^+ = \{P, S, Q, R, T\}$ (using $S \to Q$, $PQ \to R$, $R \to ST$) - **superkey!**
- All supersets of these are also superkeys.

4.2.2 (3 points)

Answer: (c)

Check BCNF: For $S \to Q$, S is not a superkey ($S^+ = \{S, Q\}$). NOT in BCNF.

Check 3NF: For $S \to Q$, Q is part of candidate key PQ, so it's acceptable in 3NF. IN 3NF.

4.2.3 (3 points)

Answer: (b)

For lossless: $(R_1 \cap R_2)^+$ must contain R_1 or R_2.

(b) $R_1 = (P, Q, R)$, $R_2 = (R, S, T)$. Common: $\{R\}$. $R^+ = \{R, S, T, Q\}$. Contains R_2! **Lossless!**

Question 5: Physical Design

5.1.1 (2 points)

Answer: (a) True

In B+ trees, internal nodes guide the search but all actual data pointers/records are in leaves. Keys may be copied up to internal nodes.

5.1.2 (2 points)

Answer: (a) True

For degree $d = 4$: max keys in leaf $= d - 1 = 3$.

5.2.1 (4 points)

Answer: (b)

After deleting 6* from [6* 8*], we have [8*] (1 key). After deleting 8*, we have [] (0 keys, severe underflow). For $d = 3$: min keys $= \lceil (3-1)/2 \rceil = 1$.

Must merge with sibling [12* 15*] to get [12* 15*]. Then restructure the tree. The leaves become: [12* 15*] | [17* 20*] | [25* 30*] Internal structure adjusts. Root has one separator.

Result: Root [17], with internal nodes [15] on left and [25] on right.

5.2.2 (4 points)

Answer: (c)

Inserting 18* into [17* 20*] makes [17* 18* 20*] (3 keys). For $d = 3$: max keys in leaf $= 2$. This overflows!

Split: [17*] and [18* 20*]. Copy up 18. New structure: Root [17], Left internal [8|12], Right internal [18|25]

Question 6: Query Processing and Optimization

6.1 (2 points)

Answer: (b)

Pages for s_1: $60,000/30 = 2,000$. 2-Way sort Phase 2 passes: $\lceil \log_2 2,000 \rceil = 11$.

6.2 (4 points)

Answer: (a), (b), (c), (d), (e)

Pages for s_2: $300,000/15 = 20,000$. (a) **Correct!**

Phase 1 I/O: read + write $= 2 \times 20,000 = 40,000$. (b) **Correct!**

Initial runs: $\lceil 20,000/250 \rceil = 80$. (c) **Correct!**

Phase 2 passes: $\lceil \log_{249} 80 \rceil = 1$. (d) **Correct!**

Total I/O: Phase 1 (40,000) + Phase 2 (1 pass $\times$ 40,000) = 80,000. (e) **Correct!**

6.3 (4 points)

Answer: (d)

Both (a) and (b) are correct. Pages: s_1 = 2,000, s_2 = 20,000. Cost with s_1 as outer: $2,000 + 2,000 \times 20,000$.

6.4 (4 points)

Answer: (a), (b), (c), (d), (e)

(a) $\sigma_{Z=100}(s)$: $5000/50 = 100$. **True!**

(b) $\sigma_{X=200}(s)$: $5000/200 = 25$. **True!**

(c) $\sigma_{Y<250}(s)$: $5000 \times \frac{250-0}{500-0} = 2500$. **True!**

(d) $\sigma_{X<200}(s)$: $5000 \times \frac{200-100}{300-100} = 5000 \times 0.5 = 2500$. **True!**

(e) $\sigma_{Z>100}(s)$: $5000 \times \frac{150-100}{150-50} = 5000 \times 0.5 = 2500$. **True!**

Question 7: Transactions, Concurrency Control, and Recovery

7.1.1 (2 points)

Answer: (a) True

REDO processes log forward (beginning to end) to reapply committed changes.

7.1.2 (2 points)

Answer: (a) True

Deadlock occurs when transactions form a cycle in the wait-for graph, each waiting for locks held by others.

7.2.1 (3 points)

Answer: (a)

Conflicts in Schedule (1):

- T2: W(M) before T1: R(M) → T2 → T1
- T1: W(N) before T3: R(N) → T1 → T3
- T2: W(P) before T3: R(P) → T2 → T3

Graph: T2 → T1, T1 → T3, T2 → T3

7.2.2 (4 points)

Answer: (b), (c), (d)

(b) Acyclic precedence graph means conflict serializable. **True!**

(c) All conflict serializable schedules are serializable. **True!**

(d) Topological sort gives: T2, T1, T3. **True!**

7.2.3 (3 points)

Answer: (b), (c)

Conflicts in Schedule (2):

- T1: R(M) before T2: W(M) → T1 → T2
- T2: W(M) before T3: W(M) → T2 → T3
- T1: W(N) before T3: R(N) → T1 → T3
- T2: W(P) before T3: R(P) → T2 → T3

The precedence graph is: T1 → T2, T2 → T3, T1 → T3.

This is acyclic! So it IS conflict serializable and serializable. Equivalent serial order: T1, T2, T3.